

An abstract graphic featuring a complex network of interconnected nodes and lines. The nodes are represented by small circles in various colors including blue, orange, purple, yellow, and black. The lines are thin and grey, creating a dense web of connections. The background is light grey with scattered black dots, suggesting a starry sky or a data field. The overall composition is dynamic and interconnected.

METAPHOR QUALITY CONTROL

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TASK DESCRIPTION

- Train a model to classify words as metaphors or metonyms
- Use linguistic data, including word senses, from ChainNet
- Evaluate model performance using accuracy

DATASET

- chainnet_metaphor.json
- chainnet_metonymy.json

TRAINING THE MODEL (PART 1)

Using scikit-learn module

- Create a tropes dictionary (metaphor + metonymy)
- Extract features from word senses
- Create features and labels for classification
- **Feature representation:**

```
text_feature = f"{from_sense}{to_sense}"
```

TRAINING THE MODEL (PART 2)

Convert text to numeric vectors using TF-IDF:

- TF-IDF transforms text into numerical features
- Important words receive higher weights

```
vectorizer = TfidfVectorizer()
```

```
X = vectorizer.fit_transform(features)
```

TRAINING THE MODEL (PART 3)

Split data into training and testing sets:

- 90% training data
- 10% testing data

```
X_train, X_test, y_train, y_test = train_test_split(  
X, labels, test_size=0.1, random_state=32)
```

TRAINING THE MODEL (PART 4)

Train Multinomial Naive Bayes classifier

- Suitable for text classification
- Works well with TF-IDF features

```
gnb = MultinomialNB()  
gnb.fit(X_train, y_train)
```

Evaluate performance with accuracy and F1-score

```
print(classification_report(y_test, y_pred))
```

EVALUATION

Evaluate performance

- Accuracy
- Precision
- Recall
- F1-score

```
print(classification_report(y_test, y_pred))
```


RESULTS

Accuracy test: 65.79%

	precision	recall	f1-score	support
metapfor	0.65	0.81	0.73	764
metonymy	0.66	0.46	0.54	598
accuracy			0.66	1362
macro avg	0.66	0.64	0.63	1362
weighted avg	0.66	0.66	0.65	1362

CONCLUSION

The model's accuracy ranges from 61% to nearly 66%.

The model performs better on metaphors than on metonymies.

Metonymy detection remains more challenging.

Performance could be improved by:

- Using richer linguistic features
- Adding more training data
- Applying more advanced models